

SALES FORECASTING USING MLOPS

Student Details

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Project Title

Sales Forecasting using Synthetic Dataset with MLflow Experiment Tracking and Azure ML Pipeline

Objective

The objective of this project is to design and implement a complete MLOps pipeline for sales forecasting. The system uses MLflow for experiment tracking and Azure ML pipeline for automating different stages of the machine learning workflow.

Dataset Description

A synthetic dataset was generated to simulate real-world sales data. The dataset consists of 500 samples with the following features:

- **Day:** Represents time progression
- **Promo:** Indicates whether a promotion is active (0 or 1)
- **Price:** Product price
- **Sales:** Target variable representing sales

The dataset is generated using NumPy to ensure realistic patterns and avoid data leakage.

Methodology

1. Data Preparation

- Synthetic dataset generated using Python
 - Stored in CSV format (sales.csv)
 - Ensures no missing values and consistent structure
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2. Training Pipeline

- Data split into training and testing sets using `train_test_split`
- Ratio used: 80% training and 20% testing
- Split datasets saved for further use

3. Model Training

Two machine learning models were used:

- **Linear Regression**
- **Random Forest Regressor**

These models were trained using different random seeds to generate multiple experiment runs.

4. Experiment Tracking (MLflow)

MLflow was used to track experiments with the following details:

- **Experiment Name:**
SKCT_727823TUAM025_SalesForecasting
 - **Number of Runs:**
12+ runs
 - **Parameters Logged:**
 - Random seed
 - **Metrics Logged:**
 - R^2 Score
 - RMSE
 - MAE
 - **Operational Metrics:**
 - Training time
 - Model size
 - **Artifacts:**
 - Best model saved
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5. Model Evaluation

The model was evaluated using standard regression metrics:

- **R^2 Score:** Measures model accuracy
- **RMSE:** Root Mean Squared Error
- **MAE:** Mean Absolute Error

Results

Metric Value

R² Score 0.6968

RMSE 17.78

MAE 13.30

Best Model

The **Random Forest Regressor** performed better compared to Linear Regression because:

- It captures non-linear relationships
- It reduces overfitting through ensemble learning
- It provides better generalization

Challenges Faced

Initially, the model produced unrealistic results:

R2: 1.0

RMSE: 0.0

Problem

This occurred due to **data leakage**, where the same dataset was used for both training and testing.

Solution

- Implemented proper `train_test_split`
- Ensured that predictions were made only on unseen test data
- Revalidated the pipeline

After fixing the issue, realistic results were obtained.

Pipeline Implementation

The project pipeline consists of three main stages:

1. **Data Preparation (`data_prep.py`)**
2. **Training Pipeline (`train_pipeline.py`)**
3. **Evaluation (`evaluate.py`)**

These stages are defined in a YAML file for automation using Azure ML pipeline.

MLflow Integration

MLflow was used to:

- Track multiple experiments
- Compare model performance
- Store trained models
- Log metrics and parameters

The MLflow UI provides a clear visualization of runs and helps in selecting the best model.

Conclusion

This project successfully demonstrates:

- End-to-end machine learning workflow
- Experiment tracking using MLflow
- Model comparison and evaluation
- Pipeline automation using YAML
- Handling and fixing real-world issues like data leakage

The final model achieved a realistic performance with an R^2 score of approximately 0.69, indicating a reliable and generalizable model.

Future Enhancements

- Use real-world datasets
 - Hyperparameter tuning
 - Deploy model using REST API
 - Integrate CI/CD pipelines
 - Use advanced models like XGBoost
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